

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Leo goes to a packing store to buy containers and tape. Leo has **\$15**. Each container costs **\$1.87** and each roll of tape costs **\$2.40**. Which inequality represents the relationship between the number of containers, c , and the number of rolls of tape, t , Leo can buy?

Answer

- A. $1.87c + 2.40t \leq 15$
- B. $1.87c + 2.40t \geq 15$
- C. $2.40c + 1.87t \leq 15$
- D. $2.40c + 1.87t \geq 15$

Correct Answer: A

Rationale

Choice A is correct. It's given that each container costs **\$1.87** and c represents the number of containers. It follows that $1.87c$ represents the cost, in dollars, of c containers. It's also given that each roll of tape costs **\$2.40** and t represents the number of rolls of tape. It follows that $2.40t$ represents the cost, in dollars, of t rolls of tape. Therefore, the total cost, in dollars, for c containers and t rolls of tape is $1.87c + 2.40t$. It's given that Leo has **\$15**. It follows that the total cost, in dollars, must be less than or equal to **15**. Therefore, the inequality $1.87c + 2.40t \leq 15$ represents this relationship.

Choice B is incorrect. This inequality represents a relationship where Leo has at least **\$15**, not at most **\$15**.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.